11 **(a)** Figure 11.1 shows some statements about the Earth and the approximate time taken for the events described in these statements to take place.

statement	approximate time taken
The Earth orbits the Sun once.	8.3 minutes
The Earth rotates once on its axis.	24 hours
Light travels from the Sun to the Earth.	1 month
The Moon orbits the Earth once.	365 days

Figure 11.1

Draw a line from each statement to the correct time taken. One has been completed for you. [2]

(b) The planet Jupiter orbits the Sun. The average radius of the orbit of Jupiter is 7.78×10^8 km. The average orbital speed of Jupiter is 4.70×10^4 km / h.

Show that the orbital period of Jupiter is 11.9 years.

[3]

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(c) (i) Define a 'light-year' in words.

.....

..... [1]

(ii) A distant galaxy is 1.6×10^{8} light-years away from the Earth.

The current estimate for the Hubble constant is 2.2×10^{-18} per second.

Calculate the speed at which this galaxy moves away from the Earth.

Give your answer in m/s.

speed = ... m/s [2]

[Total: 8]

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